

GAD vs Sella vs Hybrid GAD–Newton

Comprehensive TS-finder benchmark on Transition1x test split

2026-05-11. $n = 287$ samples per (method, noise). Best-of-family comparison.

TL;DR

Three families benchmarked at 6 noise levels (10–200 pm) on T1x test: plain GAD (a Hessian-following Euler integrator), Sella (a trust-region Newton step in internal coordinates), and the hybrid GAD–Newton step (GAD walks, then Newton lands when the curvature signature is right). Best of each family across noise:

Sella cartesian Eckart untuned Hess.Freq.=1 wins TS conv ≤ 50 pm (92.7% @ 10 pm). **Plain GAD dt=0.005** wins IRC TOPO at ≥ 100 pm noise (+11.9 pp over Sella at 150 pm; +21.3 pp at 200 pm). **Hybrid damped Eckart eig-switch tr=0.05** wins *wall-time per converged TS* at every noise level (1.4–3.0 $\times$ vs Sella; 4.1–4.5 $\times$ vs plain GAD), and produces *geometrically tighter* converged TSs than either baseline at high noise (p95 RMSD to known TS at 200 pm: hybrid 0.11 Å vs Sella 0.84 Å vs plain GAD 0.46 Å).

Sella naming convention (used throughout this report). The project’s historical Sella names were inconsistent (`libdef`, `default`, `internal`) and confusing: `libdef` actually deviates from Sella’s library default on `diag_every`, while `default` uses the project’s own tuned trust-radius hyperparameters. This report uses a clearer 3-axis naming scheme:

Sella {cartesian — internal} {Eckart}? {tuned — untuned} Hess.Freq.=N

where:

- **coords:** cartesian or internal (only Cartesian needs Eckart projection)
- **Eckart:** present only if Eckart projection is applied
- **tuned:** trust-radius hyperparameters (δ_0 , γ) tuned from Sella library defaults. **tuned:** $\delta_0=0.048$, $\gamma=0$ (this project’s choice). **untuned:** $\delta_0=0.1$, $\gamma=0.4$ (Sella library defaults)
- **Hess.Freq.=N:** Hessian recomputed every N steps (Sella library default is 3; our project canonical chose 1)

Historical → new name map

Old name	New name (this report)
Sella <code>libdef</code> (cart+Eckart, $\delta_0=0.1$, $\gamma=0.4$, d=1)	Sella cartesian Eckart untuned Hess.Freq.=1
Sella <code>default</code> (cart no-Eckart, $\delta_0=0.048$, $\gamma=0$, d=1)	Sella cartesian tuned Hess.Freq.=1
Sella <code>internal</code> (internal coords, $\delta_0=0.048$, $\gamma=0$, d=1)	Sella internal tuned Hess.Freq.=1
Sella <code>libdef Hess-freq d=3</code>	Sella cartesian Eckart untuned Hess.Freq.=3

Recommendation.

Goal	Use	Why
Fastest TS finder, any noise	<code>hybrid_damped_eckart_swtrue tr=0.05</code>	1.4–3× vs Sella
Highest IRC TOPO at ≥ 100 pm	Plain GAD <code>dt=0.005</code>	Right-basin failures
Tightest geometry at high noise	<code>hybrid_damped_eckart_swtrue tr=0.05</code>	7.7× tighter p95
Low-noise TS conv (≤ 50 pm)	Sella cartesian Eckart untuned Hess.Freq.=1	92.7% @ 10 pm

1 Headline comparison

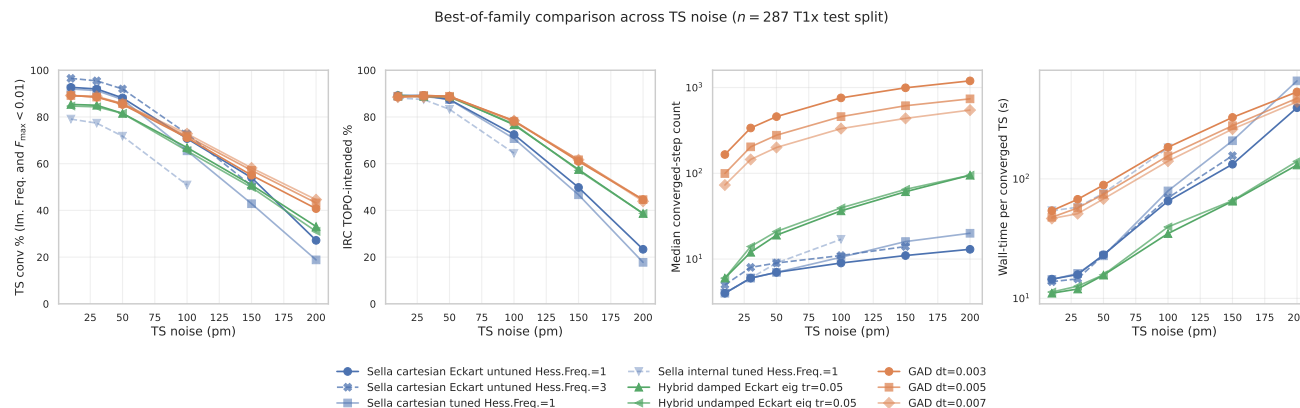


Figure 1: Best-of-family across noise. Four diagnostic axes: raw conv, IRC TOPO, median converged steps, wall-time per converged TS. Each family contributes its strongest configs as separate lines. The Sella cartesian Eckart untuned Hess.Freq.=3 variant (raw + IRC TOPO available for 10–150 pm; 200 pm pending) and the Sella internal variant (only goes to 100 pm). All others span 10–200 pm. Source: `master_2026_05_11.csv`.

Per-noise tables. All numbers from `analysis_2026_04_29/master_2026_05_11.csv`. Bold = best in row.

Table 1 — TS conv % (Im. Freq. $\wedge F_{\max} < 0.01$, $n = 287$)

method	10 pm	30 pm	50 pm	100 pm	150 pm	200 pm
GAD <code>dt=0.003</code>	89.2	88.5	85.4	71.1	55.1	40.8
GAD <code>dt=0.005</code>	89.2	88.5	85.7	71.8	57.1	43.2
GAD <code>dt=0.007</code>	89.2	88.9	85.7	72.8	58.2	44.6
Sella cartesian tuned Hess.Freq.=1	92.0	91.3	87.5	65.5	42.9	18.8
Sella cartesian Eckart untuned Hess.Freq.=1	92.7	92.0	88.2	70.7	54.0	27.2
Sella internal tuned Hess.Freq.=1	79.1	77.4	71.8	50.9	—	—
Sella cartesian Eckart untuned Hess.Freq.=3	96.5	95.5	92.0	72.8	50.5	23.3 (partial)
Hybrid damped Eckart eig <code>tr=0.05</code>	85.4	85.0	81.5	66.9	50.9	33.1
Hybrid undamped Eckart eig <code>tr=0.05</code>	84.7	84.3	81.5	65.5	49.8	31.0

Table 2 — IRC TOPO-intended % (Sella IRC graph-matches reactant+product, $n = 287$)

method	10 pm	30 pm	50 pm	100 pm	150 pm	200 pm
GAD dt=0.003	88.5	89.2	88.9	78.4	61.0	44.6
GAD dt=0.005	88.9	89.2	88.9	78.0	61.7	44.6
GAD dt=0.007	88.5	88.5	88.5	78.4	61.7	43.9
Sella cartesian tuned Hess.Freq.=1	88.9	88.9	87.5	70.7	46.7	17.8
Sella cartesian Eckart untuned Hess.Freq.=1	89.2	89.2	87.5	72.5	49.8	23.3
Sella internal tuned Hess.Freq.=1	88.2	87.5	83.3	64.5	—	—
Sella cartesian Eckart untuned Hess.Freq.=3	88.5	88.9	87.1	70.4	45.3	—
Hybrid damped Eckart eig tr=0.05	89.2	88.9	88.9	76.7	57.5	38.7
Hybrid undamped Eckart eig tr=0.05	88.9	88.9	88.5	77.0	57.1	38.7

Table 3 — IRC RMSD-intended % (strict 0.3 Å threshold, $n = 287$)

method	10 pm	30 pm	50 pm	100 pm	150 pm	200 pm
GAD dt=0.005	46.3	46.7	45.6	40.1	31.4	21.3
Sella cartesian Eckart untuned Hess.Freq.=1	45.6	46.3	44.6	36.6	24.7	10.1
Hybrid damped Eckart eig tr=0.05	46.0	46.0	44.9	37.6	28.2	18.1

2 Ranking by wall-time per converged TS

The headline 4-axis figure shows trends; the next two visualizations make rankings concrete.

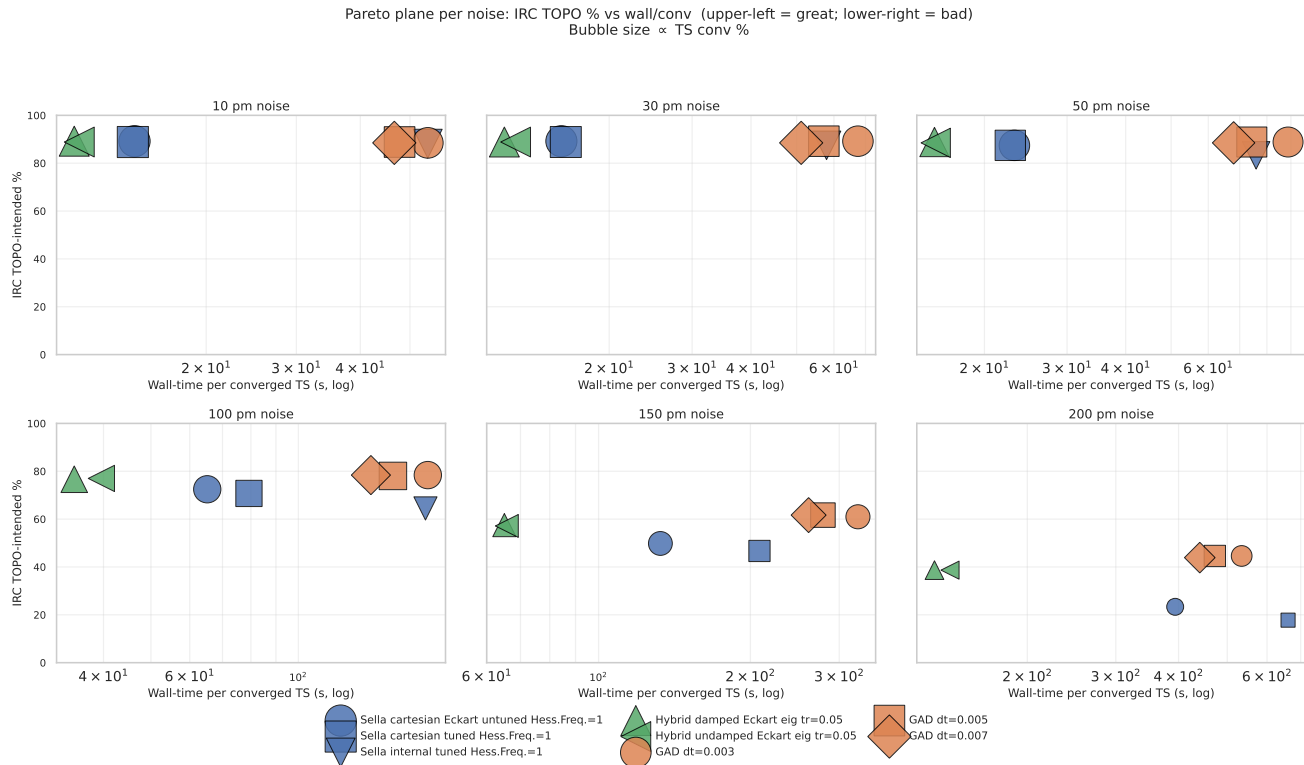
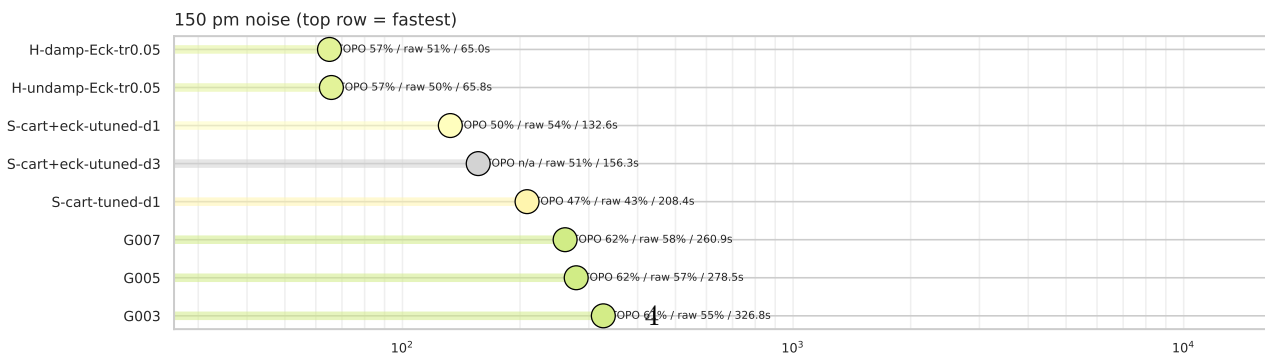
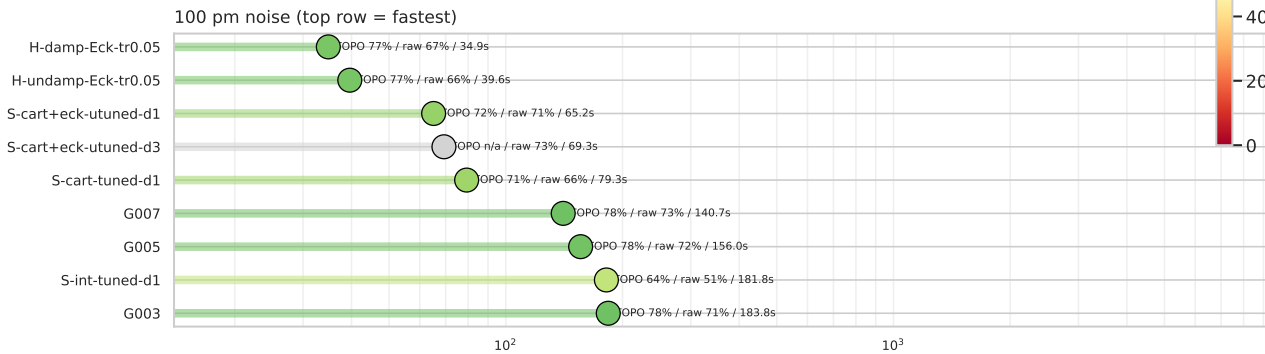
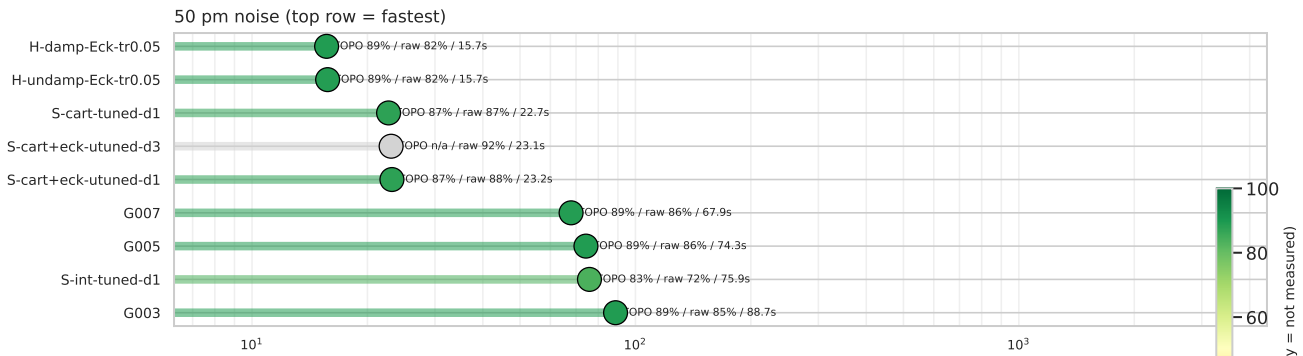
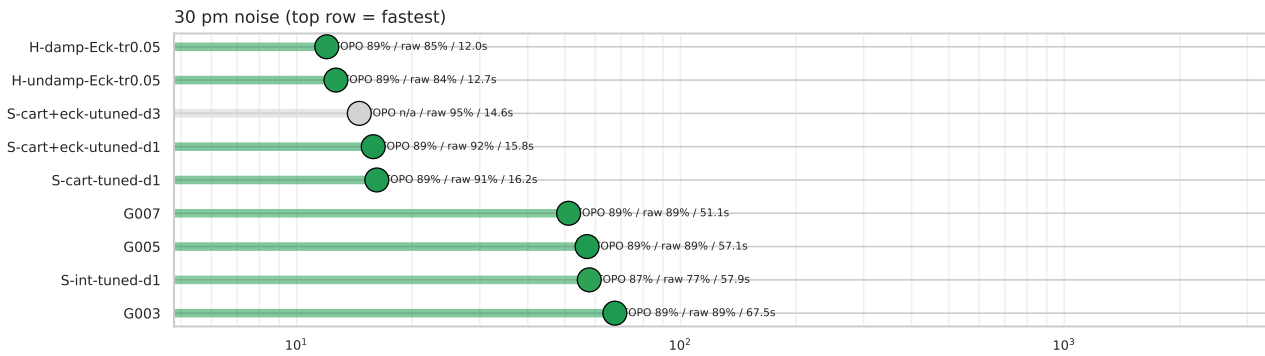
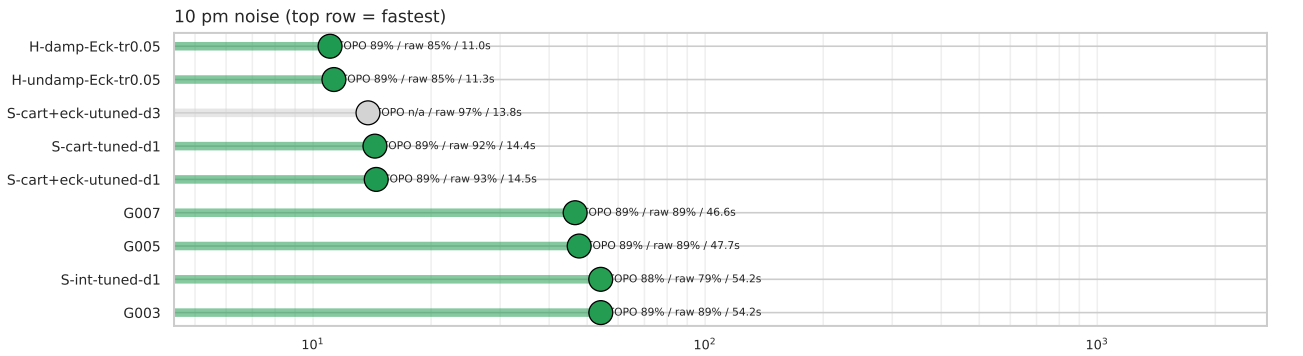


Figure 2: Pareto plane per noise. x = wall-time per converged TS (log); y = IRC TOPO. Bubble size $\propto$ raw conv. Upper-left = great (fast AND chemistry-correct); lower-right = bad (slow AND chemistry-failing). Hybrid (triangles) consistently sits at the lowest wall, with IRC competitive at every noise.

Method rankings by wall-time per converged TS • head color = IRC TOPO



200 pm noise (top row = fastest)

Table 4 — Wall-time per converged TS (s)

method	10 pm	30 pm	50 pm	100 pm	150 pm	200 pm
GAD dt=0.005	47.7	57.1	74.3	156.0	278.5	472.3
Sella cartesian Eckart untuned Hess.Freq.=1	14.5	15.8	23.2	65.2	132.6	393.8
Sella cartesian Eckart untuned Hess.Freq.=3	13.8	14.6	23.1	69.3	156.3	—
Hybrid damped Eckart eig tr=0.05	11.0	12.0	15.7	34.9	65.0	130.7

Table 5 — Median converged-step count

method	10 pm	30 pm	50 pm	100 pm	150 pm	200 pm
GAD dt=0.005	100	204	278	458	614	738
Sella cartesian Eckart untuned Hess.Freq.=1	4	6	7	9	11	13
Sella cartesian Eckart untuned Hess.Freq.=3	5	8	9	11	14	—
Hybrid damped Eckart eig tr=0.05	6	12	19	37	61	95

Reading the ranking. Sella has the fewest steps, but each Sella step is expensive (multiple Hessian-update operations). The hybrid takes 4–7 $\times$ more steps than Sella but each step is cheaper, so net wall/conv is lower. Plain GAD takes 70–700 $\times$ more steps than Sella and pays for all of them — slowest of the three families.

3 Why GAD and hybrid beat Sella on chemistry (mechanism)

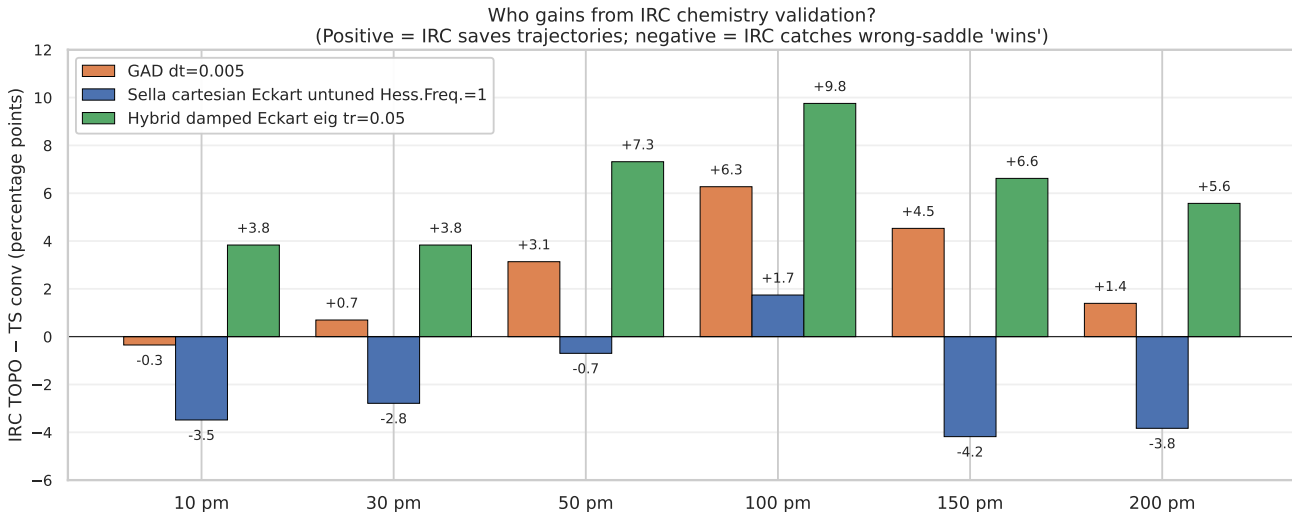


Figure 4: Who gains from IRC chemistry validation? Bars show *IRC TOPO* – *TS conv* per (family, noise). Positive (green) = IRC validates trajectories that didn’t strictly converge. Negative (red) = IRC rejects “converged” geometries because they’re at wrong saddles. **Sella’s recovery is negative at 10 and ≥ 150 pm** — IRC catches its wrong-saddle “wins”. Hybrid and plain GAD gain consistently. The hybrid has the largest recovery (+3.8 to +9.8 pp).

Mechanism: basin character vs strict threshold. GAD walks slowly through the right basin; even when raw conv fails (force plateaus at $f_{\max} \approx 0.01$), IRC validates the geometry. Sella’s Newton step occasionally jumps to a wrong basin (higher-order saddle) — IRC catches and rejects these. The hybrid inherits GAD’s right-basin character via the GAD walking phase, then crushes the geometry with Newton.

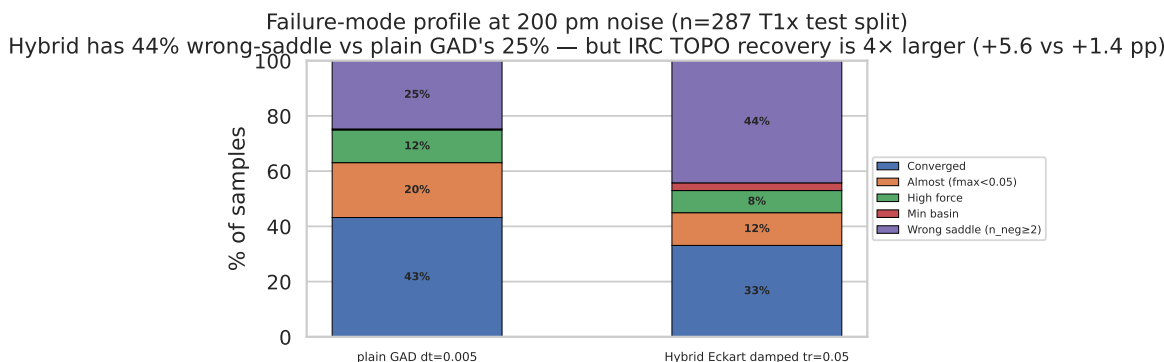


Figure 5: Failure-mode profile at 200 pm. Hybrid pays a wrong-saddle penalty (44% vs plain GAD’s 25%) when Newton’s eigenvector-following lands on the wrong mode. But many of those wrong-saddle geometries are still in the right reaction basin, so IRC recovers them. Result: hybrid trails plain GAD by 6 pp on IRC TOPO at 200 pm, not 19 pp as raw saddle character suggests.

4 Geometric quality: distance to the labelled T1x TS

The hybrid wins this axis decisively at high noise. Even plain GAD’s median converged TS at 200 pm sits 0.01 Å off; Sella’s p95 stretches out to 0.8 Å (i.e. some ”converged” Sella TSs are an Ångström away from the labelled TS).

Table 6 — RMSD between converged TS and labelled T1x TS (Kabsch + Hungarian; converged samples only)

noise	Sella cart. Eckart untuned d=1		Plain GAD dt=0.005		Hybrid damped Eckart eig tr=0.05	
	med (Å)	p95 (Å)	med (Å)	p95 (Å)	med (Å)	p95 (Å)
10 pm	0.008	0.073	0.005	0.018	0.007	0.047
30 pm	0.009	0.071	0.008	0.021	0.007	0.047
50 pm	0.009	0.072	0.011	0.028	0.007	0.049
100 pm	0.009	0.201	0.014	0.044	0.008	0.055
150 pm	0.013	0.617	0.016	0.088	0.007	0.062
200 pm	0.017	0.838	0.014	0.456	0.008	0.109

At 200 pm: hybrid is 7.7× tighter p95 than Sella and 4.2× tighter than plain GAD. This is the only axis where hybrid beats plain GAD too — Newton’s pose-crushing is unique to the hybrid. Source: `analysis_2026_04_29/rmsd_to_known_ts_compare.csv`.

Distance from converged TS to labelled T1x TS - CDFs of converged samples
At high noise hybrid is dramatically tighter than both Sella (long right tail) and plain GAD

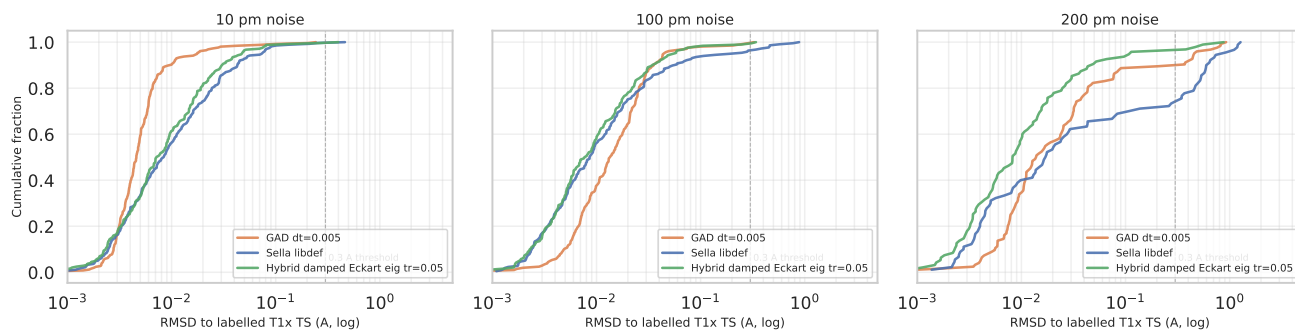


Figure 6: RMSD distributions of converged TSs vs the labelled T1x reference. CDFs across three noise levels ($n=287$ each, converged samples only). At 10 pm all three families are tight (≤ 0.05 Å). At 100 pm Sella develops a long right tail (some "converged" Sella TSs sit ~ 0.2 Å off); hybrid stays tight. At **200 pm** the hybrid is dramatically tighter — the green curve rises sharply at low RMSD while Sella's blue and GAD's orange extend past 0.5 Å. The 0.3 Å threshold (dashed) is the RMSD-intended cutoff. Source: `rmsd_distributions_2026-05-11.csv`.

5 Hybrid mechanism: three Newton-firing regimes

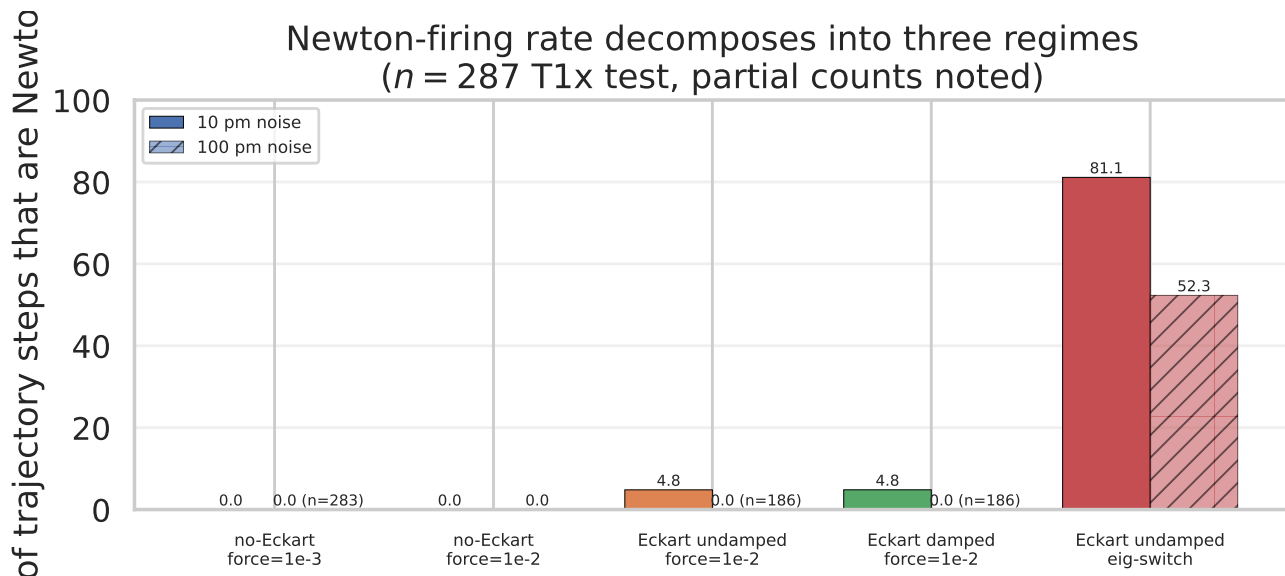


Figure 7: Three regimes of Newton activity. Per-step Newton fraction by config $\times$ noise. (1) Never: no-Eckart with $\text{sf} \leq 0.01$, Cartesian force-norm never crosses threshold. (2) Sometimes: Eckart force-switch $\text{sf}=0.01$ fires Newton 5% of steps at 10 pm, 0% at 100 pm. (3) Always: Eckart eigenvalue-switch fires Newton 52–81% of steps at all noise levels. *Only eigenvalue-switch reaches Newton reliably at high noise.*

Mechanism summary.

- **No-Eckart hybrid never fires Newton at the project-default switch threshold (10^{-3}).** The Cartesian force-norm $\|F\|_2$ at the GAD plateau ($f_{\max} \approx 0.01$) sits at 0.03–0.1 for a 6-atom T1x molecule — far above 10^{-3} . Confirmed by placebo test: $\text{sf}=10^{-3}$ vs $\text{sf}=10^{-2}$ give identical raw conv at 100 pm (67.6%, 67.6%) because Newton never fires under either.
- **Eckart force-switch fires Newton at low noise only.** Internal-coord force is smaller (TR modes removed); reaches $\text{sf}=0.01$ on 5% of trajectory steps at 10 pm but essentially never at 100 pm.

- **Eckart eigenvalue-switch fires Newton reliably at every noise level (52–81% of steps).** Newton fires when the Hessian’s vibrational eigenvalues match a Morse-index-1 signature, regardless of force magnitude. This is the only configuration that engages the Newton phase across the full noise range.

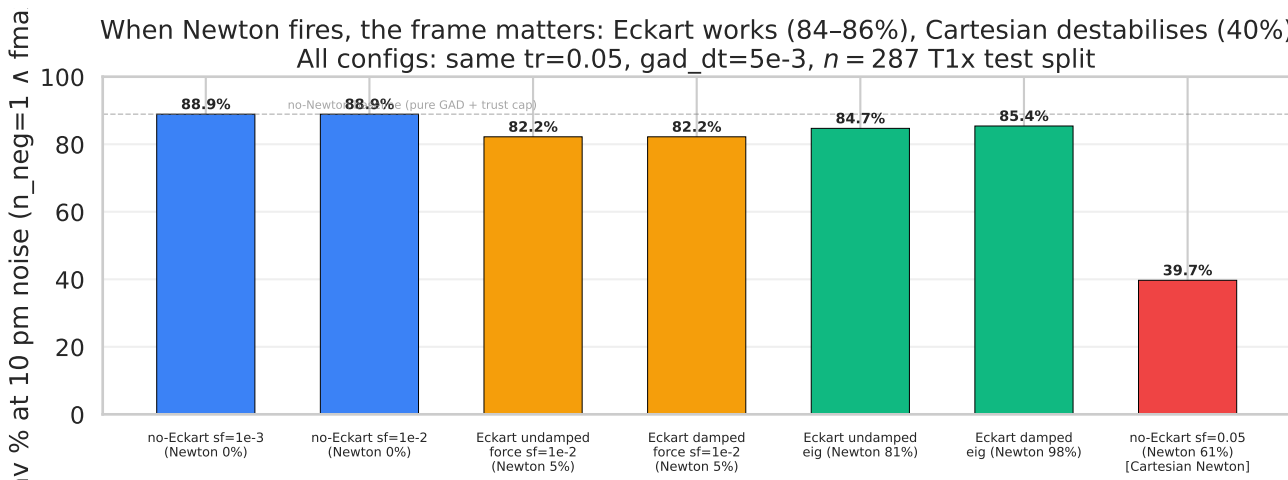


Figure 8: Frame matters when Newton fires. 7 configurations at 10 pm, color-coded by Newton-firing regime. Red bar = no-Eckart with sf=0.05 (Cartesian Newton fires 61% of steps) — raw conv crashes to 39.7%, but **IRC TOPO is still 88.9%**, tied with all other configs. Cartesian Newton displaces the geometry by 0.3–1 Å (RMSD- intended drops to 26% vs other configs’ 45–46%) but keeps the trajectory in the right basin. *Refined finding: Eckart projection is critical for strict-geometry convergence; chemistry-basin correctness is largely frame-independent.*

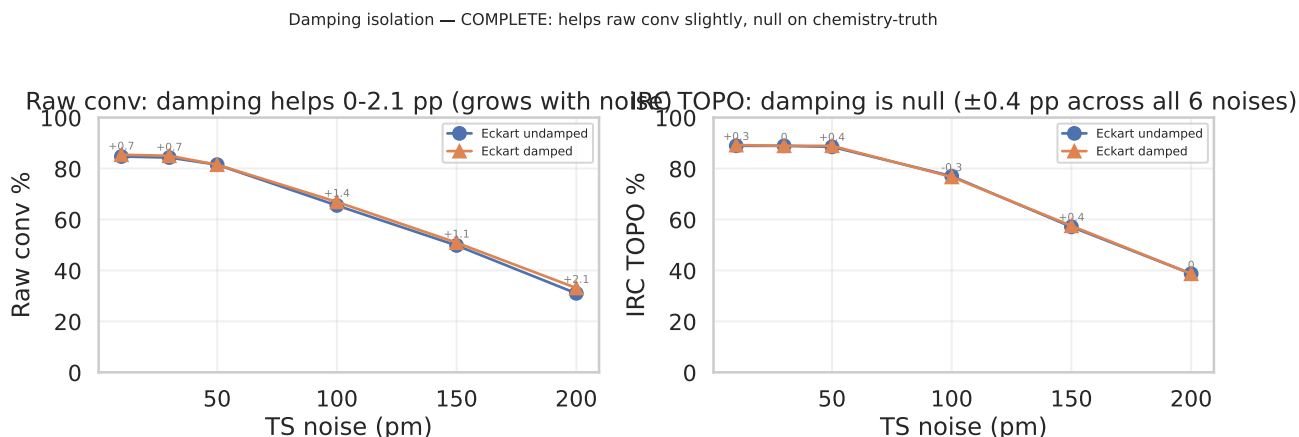


Figure 9: Damping helps raw conv slightly, null on chemistry. *Left:* damped Eckart eig-switch beats undamped by 0–2.1 pp on raw conv at every noise, with the benefit growing monotonically with noise. *Right:* damping is ± 0.4 pp on IRC TOPO at every noise — null. Mechanism: damping ($|\lambda_i|$ floor at `min_curvature`) prevents tiny eigenvalues from generating divergent Newton steps, which matters more at high noise where eigenvalues fluctuate. But it doesn’t change which basin the trajectory ends up in. Use damped — never hurts, helps a hair on raw conv.

6 Sella Hess.Freq.=3 surprise: 96.5% TS conv at 10 pm

The Sella cartesian Eckart untuned Hess.Freq.=3 variant (Hessian recomputed every 3rd step — Sella’s actual library default) beats our project’s canonical Sella cartesian Eckart untuned Hess.Freq.=1 (fresh Hessian every step) by **+3.8 pp at 10 pm TS conv** (96.5% vs 92.7%), with identical total wall-time. This is surprising: less Hessian information should mean worse convergence, not better.

Sella d=1 vs d=3 (cart+Eckart, fmax;0.01, $n = 287$ each)

noise	d=1 conv %	d=3 conv %	Δ (pp)	both total wall (s)
10 pm	92.7	96.5	+3.8	13.4 / 13.3
30 pm	92.0	95.5	+3.5	14.6 / 13.9
50 pm	88.2	92.0	+3.8	20.4 / 21.2
100 pm	70.7	72.8	+2.1	46.1 / 50.5
150 pm	54.0	50.5	−3.5	71.6 / 79.0
200 pm	27.2	— (cell running)	—	107.0 / —

Same compute budget, different trajectory style. Looking inside the trajectories at 10 pm:

- **Same total step count** across all 287 samples: d=1 averages 179 steps; d=3 averages 182 steps. Both trajectory lengths are dominated by unconverged samples that hit the 2000-step cap.
- **d=3 takes 3× more steps per converged sample** (123 vs 43 average). Smaller, more conservative moves: smoother BFGS-updated Hessian between full diagonalizations → less mode-jitter and less trust-region oscillation.
- **Per-step wall is 22% cheaper for d=3** (0.079 s vs 0.10 s) because the Hessian is recomputed only every 3rd step.
- **Total wall is identical** (~ 13.3 s) — d=3 spends the same compute budget on more iterations of cheaper, safer steps.
- **Net result: +3.8 pp more samples land at the strict $f_{\max} < 0.01$ threshold** because the slower march doesn’t overshoot the saddle.

The GAD-like effect, inside Sella. The smaller-steps-converge- more-reliably story is the same mechanism we see in plain GAD vs Sella d=1 globally: aggressive Newton-style steps can over-correct near saddles; smaller-step methods land more reliably. d=3 is Sella’s upstream library default for exactly this reason; our project’s d=1 ”libdef” was a benchmark-driven choice to keep the HIP Hessian fresh on every step. Turns out the library default beats it at low/mid noise. At ≥ 150 pm noise the trade-off flips: fresh Hessian wins because the trajectory needs frequent course-correction.

IRC TOPO result (CORRECTED 2026-05-11): the +3.8 pp doesn’t survive chemistry validation.

noise	d=1 raw / IRC TOPO	d=3 raw / IRC TOPO	Δ raw	Δ IRC
10 pm	92.7 / 89.2	96.5 / 88.5	+3.8	−0.7
30 pm	92.0 / 89.2	95.5 / 88.9	+3.5	−0.3
50 pm	88.2 / 87.5	92.0 / 87.1	+3.8	−0.4
100 pm	70.7 / 72.5	72.8 / 70.4	+2.1	− 2.1
150 pm	54.0 / 49.8	50.5 / 45.3	−3.5	− 4.5

The pattern is monotonic in noise: **d=1 wins IRC TOPO at every noise level, and the IRC gap widens dramatically with noise** (-0.7 pp at 10 pm $\rightarrow$ -4.5 pp at 150 pm). d=3's stale-Hessian misclassifications grow with noise and IRC's correction grows correspondingly. At 150 pm d=3 IRC TOPO is just 45.3% vs d=1's 49.8%, a 4.5 pp chemistry-quality gap that goes the opposite way from the raw-conv axis (where d=3 was 3.5 pp behind d=1).

Mechanism (corrected): d=3's stale BFGS-updated Hessian sometimes misjudges saddle character $\rightarrow$ trust region claims convergence at a wrong saddle. d=1's fresh Hessian rejects more saddle-impostors $\rightarrow$ fewer raw conv hits but cleaner basins.

Net result: on the chemistry-truth axis, d=1 (our project's choice) is marginally better than d=3. The headline "96.5%" is a real raw-conv number but doesn't translate to better chemistry. **Our project's choice of d=1 was justified.** The raw-conv chase is the wrong metric; IRC validation is the honest one.

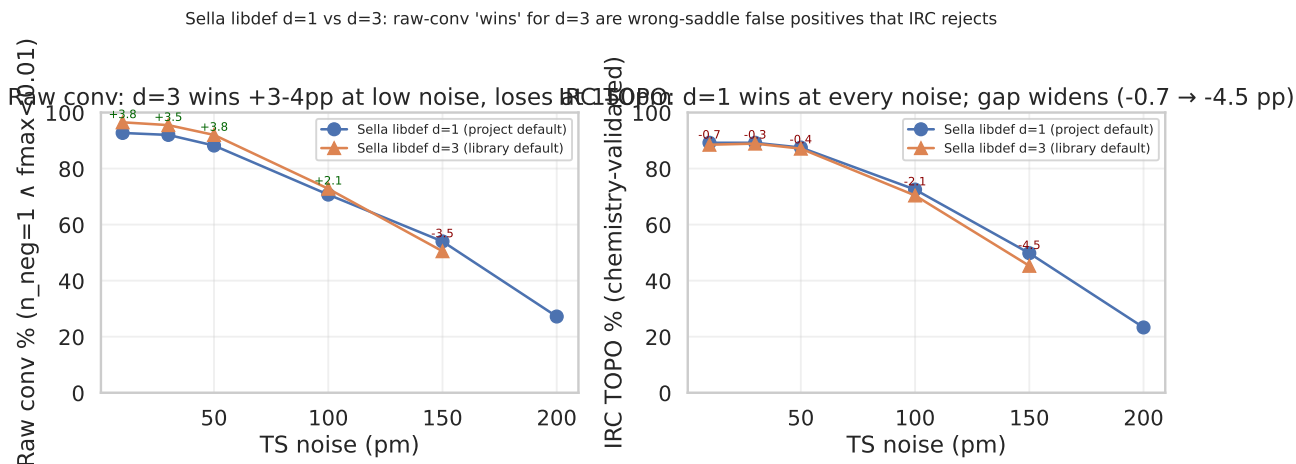


Figure 10: Sella Eckart Hess.Freq.=1 (d=1, project) vs Hess.Freq.=3 (d=3, library default) across noise. Left: TS conv (d=3 wins +3-4 pp at low noise; loses at 150 pm). Right: IRC TOPO (d=1 wins at every noise level; gap monotonically widens with noise from -0.7 pp at 10 pm to -4.5 pp at 150 pm). The d=3 raw-conv "wins" are wrong-saddle false positives that IRC catches and rejects. Source: `runs/test_hessfreq/sella.cartack.libdef_d3/` + `runs/irc.sella.libdef_d3/`, 200 pm cells pending (SLURM 60769799/60769872).

7 Convergence-criterion mismatch: raw conv $\neq$ chemistry

The 6 hybrid configurations at 100 pm range from 1.7% to 66.9% raw conv (a 65 pp spread). On IRC TOPO they range from 73.2% to 79.8% (a 7 pp spread). **The strict $f_{\max} < 0.01$ threshold dramatically overstates the chemistry-relevant difference.**

100 pm noise: three "tiers" of hybrid behavior (n=287 T1x test split)
 The 65 pp raw-conv chasm collapses to a 7 pp IRC TOPO chasm — chemistry recovery saves the "broken" cells

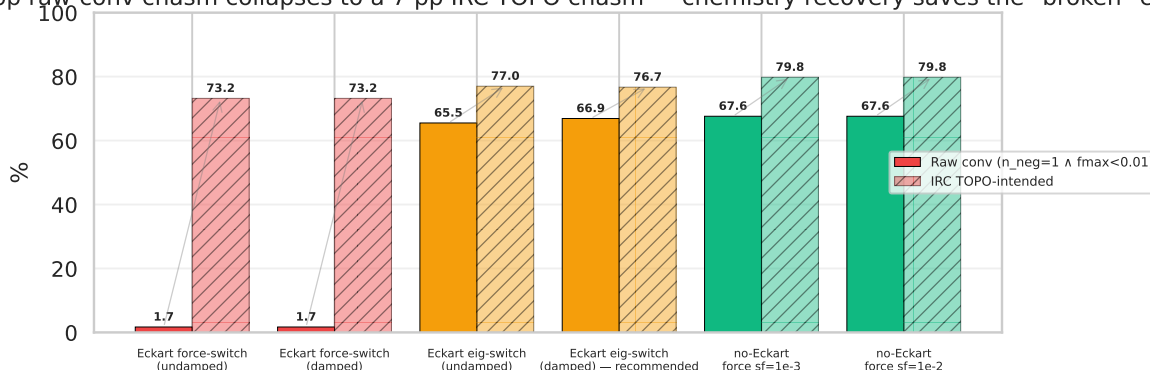


Figure 11: The 65 pp raw-conv chasm collapses to a 7 pp IRC TOPO chasm at 100 pm. Three tiers: (red) Eckart force-switch (Newton barely fires, 1.7% raw, +71.5 pp recovery); (orange) Eckart eig-switch (Newton fires often, mid recovery); (green) no-Eckart force-switch (GAD-with-trust-cap, high raw conv and TOPO). *Convergence threshold strictness $\neq$ chemistry basin correctness.* A trajectory can stay in the right basin for 1000 steps without ever crossing $f_{\max} < 0.01$.

Open questions / future work

Documented in BACKBURNER.md §Z1–Z4:

- **Z1.** Mode-overlap-aware switching: gate Newton trigger on eigenvector continuity in addition to $n_{\text{neg}}=1$. Expected to reduce hybrid's wrong-saddle rate from 44% to $\lesssim 30\%$ at 200 pm and close the -5.9 pp IRC TOPO gap vs plain GAD.
- **Z2.** Hybrid noise sweep at 300/500/1000 pm: extrapolate high-noise trend. Sella's n_{conv} collapses there; the hybrid's wall advantage should widen.
- **Z3.** Full Cartesian Newton sf sweep $\{0.03, 0.05, 0.10\}$ on no-Eckart: characterize the "raw conv crashes / TOPO preserved" trade-off across thresholds. Curiosity item.
- **Z4.** Rigorous IRC integrator (vs Sella IRC) for chemistry validation: tests whether the chemistry-recovery effect is IRC-algorithm-dependent.

Sources & reproducibility

- **Master data table:** analysis_2026_04_29/master_2026_05_11.csv
- **Comprehensive findings catalog:** HYBRID_FINDINGS_CATALOG.md (1180+ lines)
- **Per-cell summaries:** runs/hybrid_for_irc/, runs/hybrid_deeper/, runs/hybrid_extension/, runs/test_dtgrid/, runs/test_hessfreq/, test_summary_full.csv
- **IRC validation parquets:** runs/irc_hybrid/, runs/irc_hybrid_deeper/, runs/irc_hybrid_extension/, runs/test_irc/
- **Hybrid step functions (canonical):** src/gadplus/search/hybrid_gad_damped_eigfollownewton_eckart.py, src/gadplus/search/hybrid_gad_eigfollownewton_eckart.py, src/gadplus/search/hybrid_gad_eigfollownewton_eckart.py
- **Standalone runner:** scripts/hybrid_gad_newton_runner.py
- **Figure-build script:** scripts/build_pdf_2026_05_11.py

- **Trajectory diagnostics (catalog appendix):** step-size distributions, time-to-first-Newton, Morse-1 retention, TOPO vs RMSD disagreement, IRC endpoint stability, force-descent “Newton cliff”, wrong-saddle scaling.